



T.C.
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FACULTY OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

CSE4088 Introduction to Machine Learning
Term Project Proposal

Project Title

Email Spam Detection

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1. Aim of the Project

This project aims to create a dependable machine learning model to precisely categorize email communications as spam or legitimate. This classifier will assist users in eliminating undesirable and potentially possibly harmful emails, enhancing inbox organization and security. The project aims to develop a high-accuracy spam detection system utilizing a dataset from Kaggle, minimizing false positives and false negatives to enable users to concentrate on critical communications while protecting against fraudulent and dangerous content.

2. Methodology

2.1. Data Preprocessing and Exploration

Data Collection: The email dataset will be sourced from Kaggle, utilizing high-quality datasets that include diverse messages categorized as spam and legal. This dataset will serve as an appropriate foundation for the model to detect undesired (spam) email messages.

Data Cleaning: This study will deal with and correct missing and erroneous data throughout the data pre-processing phase. Missing values will be addressed by appropriate approaches, either by imputation or removal, to enhance model performance.

Feature Extraction: To enable the model to comprehend the text, feature extraction will be conducted utilizing TF-IDF (Term Frequency-Inverse Document Frequency) or Bag of Words methodologies. These strategies will transform textual data into significant features that will enhance the model's learning.

2.2. Model Development

Algorithm Selection: Appropriate algorithms will be chosen to categorize email communications as spam or legitimate. Preferred techniques include classic machine learning methods such as Logistic Regression, Naive Bayes, and Support Vector Machines (SVM).

Model Training and Testing Phases: The data will be divided into training and test sets and the performance of the model will be evaluated with the test set. By applying cross-validation, the overall performance and robustness of the model will be increased.

2.3. Model Evaluation and Optimization

Performance Metrics: Performance metrics including model accuracy, F1 score, precision, and recall will be utilized. These metrics are essential for assessing the model's efficiency in identifying spam.

Model Optimization: The model will be improved to get the best results by the adjustment of hyperparameter configurations. The model displaying optimal performance will be chosen based on comparisons performed with various methods and configurations.

3. Dataset Overview

3.1. Dataset Details

Enron Spam Dataset [1] is a dataset that is collected by V. Metsis, I. Androutsopoulos and G. Paliouras and contains spam and non-spam(ham) e-mail messages. This dataset is recorded in such a way, that every single mail is in a separate txt file, distributed over several directories. The ham(non-spam) messages are recorded under the following collections: “beck-s.tar”, “farmer-d.tar”, “kaminski-v.tar”, “kitchen-l.tar”, “lokey-m.tar” and “williams-w3.tar”. Similarly, the spam messages are under: “BG.tar”, “GP.tar” and “SH.tar”.

3.2. Dataset Overview Table

This section provides an overview of the Enron Spam Dataset, detailing key information such as its source, description, and the distribution of spam and non-spam email messages. It includes the total number of instances in the dataset, with a breakdown of spam and non-spam messages.

Dataset Name	Source	URL	Description	Number of Spam	Number of Non-Spam	Total Number of Instances
Enron Spam Dataset	Open Source	Link	This dataset contains spam and non-spam e-mail messages.	17.171	16.545	33.716

Table 3.2.1 Overview of the Enron Spam Dataset

3.3. Dataset Attributes

This section outlines the key attributes of the Enron Spam Dataset, providing details about each attribute's name, type, and explanation.

Attribute Name	Attribute Type	Explanation
Subject	string	The subject line of the e-mail
Message	string	The content of the e-mail
Date	datetime	The date the e-mail arrived. It has a YYYY-MM-DD format.
Label	category	Indicates whether the message was categorized as “spam” or “ham”.

Table. 3.3.1 Dataset Attributes for the Enron Spam Dataset

4. Draft Time Plan

4.1. Member Roles and Responsibilities

The table below details each team member's role and responsibilities, ensuring clear task allocation, accountability, and effective collaboration throughout the project.

Member	Role	Responsibilities
Emir Büçkün	Team Leader	Ensures effective organization and collaboration within the group.
Fatma Melisa Küçük	Feature Engineering	Identifies and extracts relevant features from the dataset.
Ayşenur Tüfekçi	Machine Learning	Responsible for tuning hyperparameters and improving the performance of the models.
İrem Kubalas	Model Optimization	Focuses on selecting algorithms, training models, and evaluating their performance.
Mehmet Sina Çağlar	Data Engineering	Responsible for gathering, cleaning, and preparing the data, and analyzing it to identify patterns.

Table. 4.1.1 Member Roles and Responsibilities for Specific Tasks

These role descriptions apply to specific tasks only. Beyond these, all project work will be carried out collaboratively, with everyone contributing to the core activities.

4.2. Project Timeline

Below is the project timeline figure, detailing the key phases and milestones for this project.

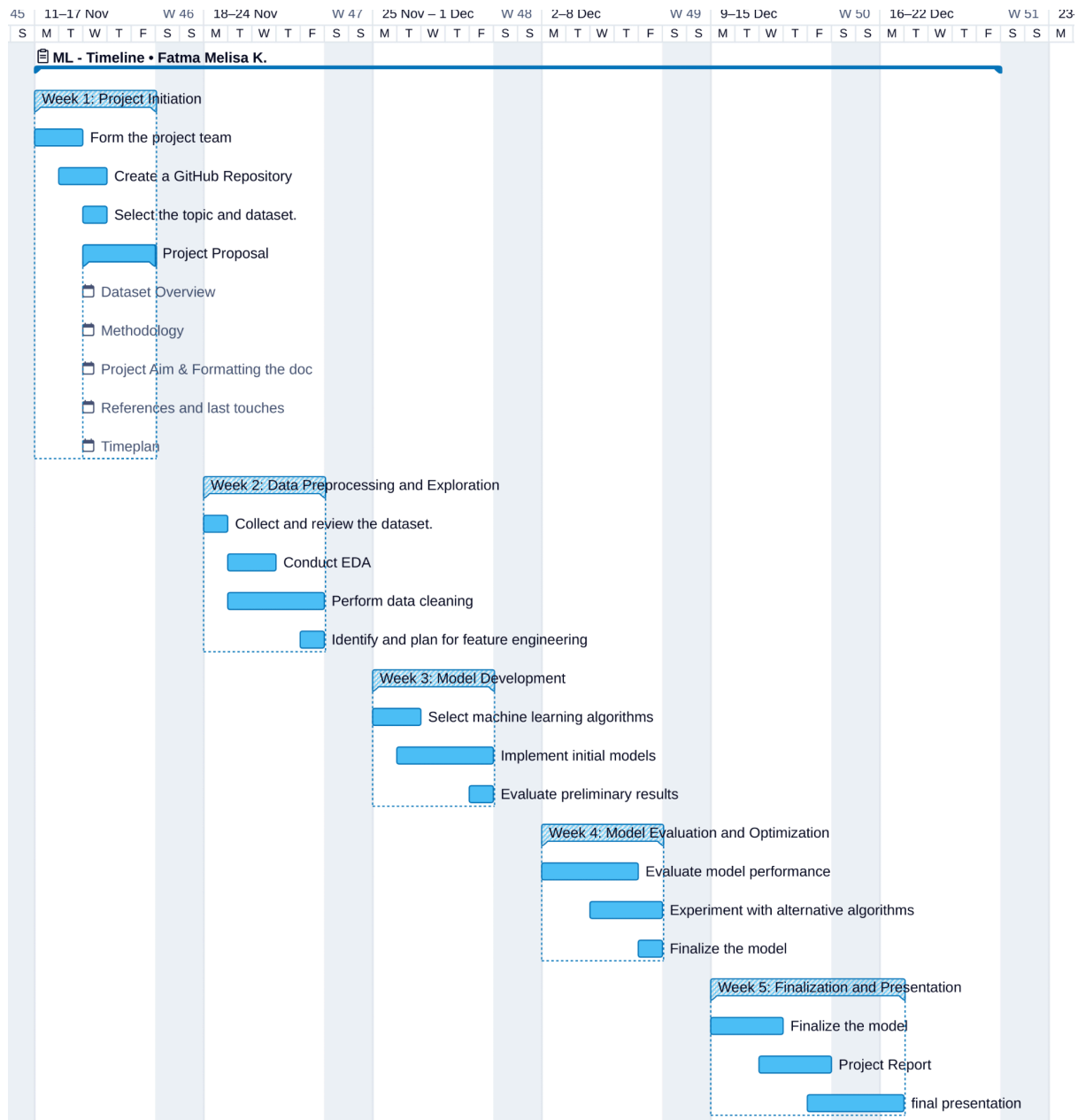


Fig. 4.2.1 Project Timeline Gantt Chart Created Using Wrike [3]

5. References

[1] V. Metsis, I. Androutsopoulos and G. Paliouras, "Spam Filtering with Naive Bayes - Which Naive Bayes?". Proceedings of the 3rd Conference on Email and Anti-Spam (CEAS 2006), Mountain View, CA, USA, 2006.

[2] <https://www.kaggle.com/datasets/wanderfj/enron-spam>

[3] Wrike - Versatile and Robust Project Management Software
<https://www.wrike.com/>